ExploratoryGenius

```
# Load in jsonlite package
library(jsonlite)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(igraph)
```

```
Attaching package: 'igraph'

The following objects are masked from 'package:dplyr':

    as_data_frame, groups, union

The following objects are masked from 'package:stats':

    decompose, spectrum
```

```
The following object is masked from 'package:base':

    union
```

```
#Assign Node Data and get a snapshot

#datausers_nominal <- stream_in(file("/Users/alexgould/Desktop/STA585GeniusProject/genius-exp
#head(datausers)
#data_annotate <- stream_in(file(("//Users/alexgould/Desktop/STA585GeniusProject/genius-exper
#head(data_annotate)
#songinfo <-  stream_in(file(("/Users/alexgould/Desktop/STA585GeniusProject/genius-expertise/
#head(songinfo)
# save(datausers_nominal, data_annotate, songinfo, file = "/Users/alexgould/Desktop/STA585Ge

# Load in Node data
load("/Users/alexgould/Desktop/STA585GeniusProject/initialfiles.Rdata")

# Assign Edge Data
edges_nominal <- read.csv("/Users/alexgould/Desktop/STA585GeniusProject/genius-expertise/foll
length(unique(edges_nominal$follower))
```

```
[1] 118199
```

```
length(unique(edges_nominal$following))
```

```
[1] 737206
```

```
#Make Dataset with only users in node dataset and who are also contained
# in folower and following

valid_users <- intersect(
  unique(data_annotate$user[!is.na(data_annotate$user)]),
  intersect(
    intersect(unique(datausers_nominal$url_name),
              unique(edges_nominal$follower)),
    unique(edges_nominal$following)
  )
)
edges_reduced <- data.frame()
edges_reduced <- edges_nominal[edges_nominal$follower %in% valid_users &
                                 edges_nominal$following %in% valid_users,
```

```
                                c("follower", "following")]
#stopifnot(all(unlist(edges_reduced) %in% nodes_reduced$url_name))
dim(edges_reduced)
```

```
[1] 15884     2
```

```
valid_nodes <- datausers_nominal[datausers_nominal$url_name %in% valid_users, ]

valid_annotate <- data_annotate[data_annotate$user %in% valid_users, ]


nodes_reduced <- merge(valid_nodes, valid_annotate, by.x = "url_name",
                       by.y = "user", all.x = TRUE)
dim(nodes_reduced)
```

```
[1] 24133    18
```

```
#Infeasible to cast to wide because of the amount of songs per user
nodes_r <- nodes_reduced %>%
            group_by(url_name) %>%
            slice_max(votes, n=1, with_ties = F)%>%
  ungroup()
#Clear empty columns
nodes_r <- as.data.frame(nodes_r[,setdiff(names(nodes_r), c("acceptor", "artist"))])
#Change classes of vectors
nodes_r <- cbind(nodes_r[, names(nodes_r) != "stats"], nodes_r$stats)

g <- graph_from_data_frame(d= edges_reduced, directed = T, vertices = nodes_r)
```